Least-Squares Filter Design

ERTH 467 — Geoscientific Signal Processing

From prediction error to the normal equations

1 What problem are we trying to solve?

Suppose we know an input signal s and a desired output (or *target*) signal y. We seek the coefficients of a linear filter f such that convolving s with f produces an output as close as possible to y:

$$\widehat{y}_n = \sum_{k=0}^{M-1} f_k \, s_{n-k}. \tag{1}$$

Here M is the number of unknown filter coefficients. We collect samples into vectors, and write the convolution as a matrix multiplication:

$$\boxed{\widehat{\mathbf{y}} = \mathbf{S}\mathbf{f}.} \tag{2}$$

For N output samples, $\mathbf{S}$ is an $N \times M$ matrix, $\mathbf{f}$ is an $M \times 1$ vector, and $\mathbf{y}$ and $\widehat{\mathbf{y}}$ are $N \times 1$ vectors. Each column of $\mathbf{S}$ is a shifted copy of the input signal s.

Matrix convention. For finite, zero-padded, causal convolution, $\mathbf{S}$ has Toeplitz structure, for example

$$\mathbf{S} = \begin{bmatrix} s_0 & 0 & 0 \\ s_1 & s_0 & 0 \\ s_2 & s_1 & s_0 \\ s_3 & s_2 & s_1 \\ \vdots & \vdots & \vdots \end{bmatrix}. \tag{3}$$

This is the type of matrix displayed in the lecture slide. If we instead assume *periodic* indexing, s_{n-k} is interpreted modulo N and $\mathbf{S}$ can be a *circulant* matrix. For instance, with $N = M = 3$,

$$\mathbf{S}_{\text{circ}} = \begin{bmatrix} s_0 & s_2 & s_1 \\ s_1 & s_0 & s_2 \\ s_2 & s_1 & s_0 \end{bmatrix}. \tag{4}$$

The least-squares derivation below is identical in either case.

2 Define the prediction error

For each sample, define the prediction error as the difference between the observed target and the filtered signal:

$$e_n = y_n - \widehat{y}_n = y_n - \sum_{k=0}^{M-1} S_{nk} f_k. \tag{5}$$

In vector form,

$$\boxed{\mathbf{e} = \mathbf{y} - \mathbf{S}\mathbf{f}.} \tag{6}$$

There are generally more data samples than unknown filter coefficients ($N \geq M$), so it may not be possible to make every e_n exactly zero. Instead, choose $\mathbf{f}$ to minimize the *sum of squared prediction errors*:

$$\boxed{J(\mathbf{f}) = \sum_{n=0}^{N-1} e_n^2 = \mathbf{e}^T\mathbf{e} = (\mathbf{y} - \mathbf{S}\mathbf{f})^T(\mathbf{y} - \mathbf{S}\mathbf{f}).} \tag{7}$$

This is the least-squares objective. It is a *scalar* function of the M unknown coefficients $f_0, \ldots, f_{M-1}$.

3 Find the minimum by differentiating

The key idea is the same as in elementary calculus: at a differentiable minimum, the derivative with respect to each adjustable coefficient is zero. With several unknowns, we collect all partial derivatives in the *gradient*:

$$\nabla_{\mathbf{f}} J = \begin{bmatrix} \partial J/\partial f_0 \\ \partial J/\partial f_1 \\ \vdots \\ \partial J/\partial f_{M-1} \end{bmatrix}, \qquad \text{and require} \qquad \nabla_{\mathbf{f}} J = \mathbf{0}. \tag{8}$$

Step 1: Expand the squared error

Using $(\mathbf{a} - \mathbf{b})^T = \mathbf{a}^T - \mathbf{b}^T$,

$$J(\mathbf{f}) = (\mathbf{y}^T - \mathbf{f}^T\mathbf{S}^T)(\mathbf{y} - \mathbf{S}\mathbf{f}) \tag{9}$$

$$= \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{S}\mathbf{f} - \mathbf{f}^T\mathbf{S}^T\mathbf{y} + \mathbf{f}^T\mathbf{S}^T\mathbf{S}\mathbf{f}. \tag{10}$$

The two middle terms are both *scalars* and equal because one is the transpose of the other. Therefore,

$$\boxed{J(\mathbf{f}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{f}^T\mathbf{S}^T\mathbf{y} + \mathbf{f}^T\mathbf{S}^T\mathbf{S}\mathbf{f}.} \tag{11}$$

Step 2: Differentiate with respect to the filter

Treat $\mathbf{S}$ and $\mathbf{y}$ as known; only $\mathbf{f}$ is variable. The three terms of Eq. (11) have gradients

$$\nabla_{\mathbf{f}}(\mathbf{y}^T\mathbf{y}) = \mathbf{0}, \qquad \text{(independent of } \mathbf{f}\text{)}, \tag{12}$$

$$\nabla_{\mathbf{f}}(-2\mathbf{f}^T\mathbf{S}^T\mathbf{y}) = -2\mathbf{S}^T\mathbf{y}, \qquad \text{(linear in } \mathbf{f}\text{)}, \tag{13}$$

$$\nabla_{\mathbf{f}}(\mathbf{f}^T\mathbf{S}^T\mathbf{S}\mathbf{f}) = 2\mathbf{S}^T\mathbf{S}\mathbf{f}, \qquad \text{(quadratic in } \mathbf{f}\text{)}. \tag{14}$$

The last identity holds because $\mathbf{S}^T\mathbf{S}$ is symmetric. Combining the three derivatives gives

$$\boxed{\nabla_{\mathbf{f}} J = -2\mathbf{S}^T\mathbf{y} + 2\mathbf{S}^T\mathbf{S}\mathbf{f} = -2\mathbf{S}^T(\mathbf{y} - \mathbf{S}\mathbf{f}).} \tag{15}$$

Interpretation: the derivative tells us how the total squared error changes when we slightly modify each coefficient of the filter.

Step 3: Set the gradient to zero

At a minimum,

$$\nabla_{\mathbf{f}} J = \mathbf{0}, \tag{16}$$

$$-2\mathbf{S}^T\mathbf{y} + 2\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{0}, \tag{17}$$

$$\boxed{\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}.} \tag{18}$$

These are the **normal equations**, a system of M linear equations for the M unknown filter coefficients.

If $\mathbf{S}^T\mathbf{S}$ is invertible, the formal solution is

$$\boxed{\mathbf{f}_{\mathrm{LS}} = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\mathbf{y}.} \tag{19}$$

This is the expression on the lecture slide. In an actual numerical program, we generally *solve* the linear system rather than explicitly constructing an inverse. If $\mathbf{S}$ has linearly dependent columns, the inverse does not exist; a pseudoinverse or regularization can be used instead.

4 Why does this stationary point minimize the error?

The curvature of the objective is described by its Hessian:

$$\nabla_{\mathbf{f}}^2 J = 2\mathbf{S}^T\mathbf{S}. \tag{20}$$

For any vector $\mathbf{v}$,

$$\mathbf{v}^T\mathbf{S}^T\mathbf{S}\mathbf{v} = (\mathbf{S}\mathbf{v})^T(\mathbf{S}\mathbf{v}) = \|\mathbf{S}\mathbf{v}\|_2^2 \geq 0. \tag{21}$$

Thus J is a convex quadratic function, and a solution of the normal equations is a *global minimum*, not a maximum. The minimum is unique when the columns of $\mathbf{S}$ are linearly independent.

There is also a useful geometric interpretation. From Eq. (15), at the optimum

$$\boxed{\mathbf{S}^T\mathbf{e}_{\mathrm{LS}} = \mathbf{0}.} \tag{22}$$

The remaining prediction error is orthogonal to every column of $\mathbf{S}$. In other words, no additional linear combination of the available shifted signal samples can reduce the squared error.

5 Connection with correlation

It is common to introduce

$$\mathbf{R} = \mathbf{S}^T\mathbf{S}, \qquad \mathbf{p} = \mathbf{S}^T\mathbf{y}. \tag{23}$$

Then the normal equations become

$$\boxed{\mathbf{R}\mathbf{f} = \mathbf{p}.} \tag{24}$$

For a convolution matrix, entries of $\mathbf{R}$ contain inner products of shifted copies of the input signal (related to *autocorrelation*), while entries of $\mathbf{p}$ contain inner products between shifted input samples and the target (related to *cross-correlation*). Exact edge behavior depends on whether we use finite zero-padded or periodic convolution.

6 Optional: check the derivative one coefficient at a time

For students who prefer ordinary scalar differentiation, start directly from

$$J = \sum_{n=0}^{N-1} \left(y_n - \sum_{k=0}^{M-1} S_{nk} f_k \right)^2 . \tag{25}$$

Differentiate with respect to one coefficient f_j using the chain rule:

$$\frac{\partial J}{\partial f_j} = 2 \sum_{n=0}^{N-1} \left(y_n - \sum_{k=0}^{M-1} S_{nk} f_k \right) (-S_{nj}) \tag{26}$$

$$= -2 \sum_{n=0}^{N-1} S_{nj} e_n . \tag{27}$$

Setting this equal to zero for every $j = 0, \ldots, M-1$ gives

$$\sum_{n=0}^{N-1} S_{nj} \sum_{k=0}^{M-1} S_{nk} f_k = \sum_{n=0}^{N-1} S_{nj} y_n , \tag{28}$$

which is precisely the jth row of $\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}$. The scalar derivation and the matrix derivation are therefore the same calculation.

Main result to remember

A least-squares filter chooses the coefficients $\mathbf{f}$ that make the predicted signal $\mathbf{S}\mathbf{f}$ as close as possible to the target $\mathbf{y}$ in the *sum-of-squared-errors* sense:

$$\underbrace{J(\mathbf{f}) = (\mathbf{y} - \mathbf{S}\mathbf{f})^T (\mathbf{y} - \mathbf{S}\mathbf{f})}_{\text{define error}} \quad \xrightarrow{\nabla_{\mathbf{f}} J = 0} \quad \underbrace{\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}}_{\text{solve for filter}} .$$